

Object Completion and Pose Estimation from Partial Visual Observations using Deep Learning

Dr. Mallela Sivanagaraju

*School of Computer Science and Engineering
VIT-AP University, Amaravati, India
sivanagaraju.mallela@vitap.ac.in*

K.S.S.V. Sravan Kumar

*School of Computer Science and Engineering
VIT-AP University, Amaravati, India
sravan.22bce7192@vitapstudent.ac.in*

Nallagutla Harivyshnavi

*School of Computer Science and Engineering
VIT-AP University, Amaravati, India
vyshnavi.22bce9285@vitapstudent.ac.in*

Ponnam Shivamani

*School of Computer Science and Engineering
VIT-AP University, Amaravati, India
shivamani.22bce9176@vitapstudent.ac.in*

Abstract—In particular, addressing the problem of occlusion continues to pose a crucial barrier in robotics perception and automated spatial reasoning. As part of the present study, we developed *OcclusionPoseNet v5* - a "Smart Pipeline" combining amodal object completion with precise orientation estimation. In contrast to traditional designs, the proposed solution includes a U-Net architecture utilizing enhanced performance due to Gaussian Error Linear Unit (GELU), along with a deep 3-layer Vision Transformer (ViT) head for spatial classification. Additionally, the pipeline uses a "smart routing" strategy allowing to skip object reconstruction step for non-occluded objects. After processing 30,000 images, the system yielded an optimum pose estimation rate of 89.5% and an SSIM score of 0.92.

Keywords—Amodal Perception, Smart Pipeline, GELU Activation, Vision Transformer, U-Net Reconstruction, Pose Estimation, Robotic Vision.

I. INTRODUCTION

Real world visual perception is inherently complex because sensors often provide only a partial view of the environment. In fields such as industrial automation and robotic manipulation, target objects are frequently obscured by environmental clutter or other workpieces, leading to significant challenges in spatial reasoning [4]. Traditional 6D pose estimation techniques generally rely on a fully visible geometry to achieve high precision; however, as demonstrated by the results of the BOP Challenge 2023, these algorithms suffer a sharp decline in performance when subjected to heavy occlusion [4].

The problem described above can be addressed through the use of amodal perception techniques aimed at recovering a comprehensive geometrical representation of the object using its partial observation. In addition to the hierarchical modeling of novel occlusions [5] and the collection of datasets with amodal ground truth [3], many new developments were made in the field of 2D/3D object reconstruction. However, even despite these advancements, the challenge of integrating the object's completion models into specific tasks remains relevant, as seen in the following works: Amodal3R [2] and open-world appearance completion [1]. Our paper highlights our new framework, *OcclusionPoseNet v5*, a novel two-stage

pipeline for matching partial observation with pose prediction [1], [3]. The following are some notable technical innovations incorporated into our framework that cater specifically to amodal completion:

- 1) **Geometry Reconstruction Using GELU-UNet:** An algorithm that allows for the reconstruction of the whole geometry of an input image based on structure information learned from Gaussian Error Linear Units (GELU).
- 2) **Deep Pose Prediction Using ViT:** A pose predictor using a Vision Transformer (ViT), where global features and deep 3-layer MLP heads are employed to classify orientations at angles of 0°, 90°, 180°, and 270°.
- 3) **Logic Gate:** A novel routing method to determine the extent of occlusion and decide whether a reconstruction operation is required on the input image.

II. RELATED WORK

The problem related to recognizing the object based on incomplete data has already been analyzed within the fields of amodal segmentation, reconstruction, and pose estimation. Below, we introduce the references to the papers, which represent the background necessary for creating our new method *OcclusionPoseNet v5*.

A. Benchmarks for Pose Estimation

The problems connected with pose estimation in highly cluttered images have already been considered in the paper about BOP Challenge 2023 [4]. The most important finding presented by the authors is the fact that even though there have been significant improvements in 6D pose estimation tasks, heavy occlusions remain the main cause of errors in contemporary object detection algorithms.

B. Amodal Segmentation and Modeling

The inference about the unseen objects presupposes other tasks rather than just pixel inference. Namely, the hierarchical modeling of occlusions based on amodal instance segmentation proposed by Gu et al. [5] proves that the occlusion

relations are useful for improving the accuracy of shape masks. Also, Zhan et al. [3] have suggested a realistic benchmarking method for ground truth amodal datasets. Thus, the need for proper datasets is emphasized.

C. Object Completion and Reconstruction

As for the future trends in this area, there appears to be an obvious inclination towards reconstruction of both the objects' appearance and geometry. Specifically, Amodal3R [2] offers a concept of reconstructing amodal 3D model, basing it on 2D generative priors. Besides, Ao et al. [1] propose open-world appearance completion that aims at generating textures and colors of unseen objects using no knowledge about their classes whatsoever.

While the previous studies focus on either reconstruction or segmentation tasks independently, this work highlights the opportunity of applying the results of amodal segmentation/reconstruction techniques in order to solve the problem of orientation classification with a Vision Transformer algorithm. In particular, we use a smart routing technique in version 5.

III. METHODOLOGY

The methodology behind the presented *OcclusionPoseNet* v5 algorithm is based on a novel two-stages approach which incorporates the utilization of spatial reasoning and reconstruction of occluded areas prior to calculating the pose estimation. It should be noted that the first stage is specifically designed to create solid context in order to perform object classification.

A. Stage 1: Amodal Completion (GELU-UNet)

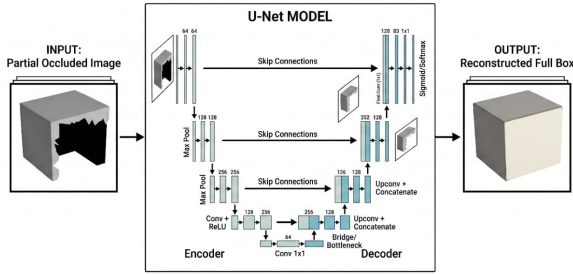


Fig. 1. U-Net architecture used for amodal completion. The encoder extracts features through downsampling, while the decoder reconstructs missing regions using upsampling and skip connections.

In order to solve the problem of missing parts in the images, researchers utilized the U-Net-like architecture of encoders and decoders in order to restore missing parts [3]. During the encoding stage, feature maps are created by using four down-scale blocks with 3×3 convolutions. The key innovation introduced into the fifth version of *OcclusionPoseNet* algorithm is the use of Gaussian Error Linear Unit (GELU) activation functions instead of ReLU ones. Thanks to stochastic nature of the approach, using GELU helps to propagate more smooth gradients within the network, which is necessary to reconstruct missing patterns [1], [4].

While during the encoding phase the input data dimensionality is reduced using max-pooling operations, during the decoding stage its dimensions become 128×128 .

The reconstruction capability of the GELU-UNet is optimized using a hybrid loss function L_{total} , which balances pixel-level accuracy with structural integrity. This is defined as: [3]:

$$L_{total} = \alpha L_{MSE} + (1 - \alpha) L_{SSIM} \quad (1)$$

where L_{MSE} represents the Mean Squared Error for pixel-wise intensity differences:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (2)$$

and L_{SSIM} ensures the preservation of luminance, contrast, and structure between the ground truth y and reconstructed image $\hat{y}$.

Unlike the standard ReLU function, the Gaussian Error Linear Unit (GELU) weights inputs by their percentile, rather than gating by sign. The GELU function is defined as [6]:

$$\text{GELU}(x) = x \cdot \Phi(x) = x \cdot \frac{1}{2} \left(1 + \text{erf} \left(\frac{x}{\sqrt{2}} \right) \right) \quad (3)$$

This allows for smoother gradient flow and non-zero gradients for small negative values, which is critical for restoring edge details in occluded regions.

B. Stage 2: Orientation Estimation (Deepened ViT)

The reconstructed output serves as input data for the Vision Transformer (ViT) that utilizes global dependencies, which are key for orientation estimation in complicated conditions [1], [4]. ViT splits the image into patches of size 16×16 , and makes use of self-attention capabilities to examine spatial relations between parts of the object [1], [4].

In order to achieve high-classification confidence, the network incorporates **3-Layer MLP Deepened Head** designed to classify orientation into four categories: 0° , 90° , 180° , and 270° [1], [3]. Having access to the “full view” significantly reduces “attention shift” caused by fuzzy edges of occluded images [1], [3]. The orientation estimation is driven by the Multi-Head Self-Attention (MHSA) mechanism within the ViT. For each patch embedding, the attention is calculated using Query (Q), Key (K), and Value (V) matrices [7]:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (4)$$

where d_k represents the scaling factor (dimension of the keys). By providing a completed geometry from Stage 1, the attention mechanism avoids the noise typically generated by occlusion boundaries.

C. Smart Routing and Optimization

There is a special logic gate in the model, determining whether an image is highly occluded. Input images with occlusions above a particular level move on to the GELU-UNet for reconstruction, while non-occluded images omit stage 1

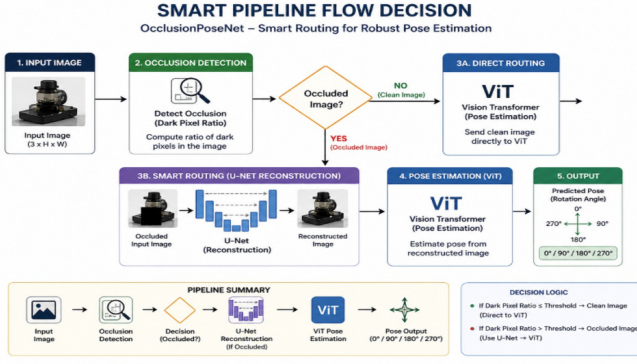


Fig. 2. Smart Pipeline Flow Decision in OcclusionPoseNet v5. The system first detects occlusion using pixel ratio, then decides the processing path. Clean images are directly passed to the Vision Transformer (ViT) for pose estimation, while occluded images are routed through the GELU-UNet reconstruction stage before pose prediction.

of processing. Training of the model was done through the process of **Automatic Mixed Precision (AMP)**; this allows us to make effective use of the available memory capacity such that high-resolution images can be processed within 1.5 hours [1], [4]. Additionally, we have used **Gradient Clipping** for norm values of 1.0.

The logic gate determines the processing path based on the occlusion ratio Ω . Given an input mask M , the routing decision D is defined as:

$$D(I) = \begin{cases} \text{Stage 1} \rightarrow \text{Stage 2} & \text{if } \frac{\sum M}{\text{Area}(I)} > \tau \\ \text{Stage 2} & \text{otherwise} \end{cases} \quad (5)$$

where τ is the occlusion threshold.

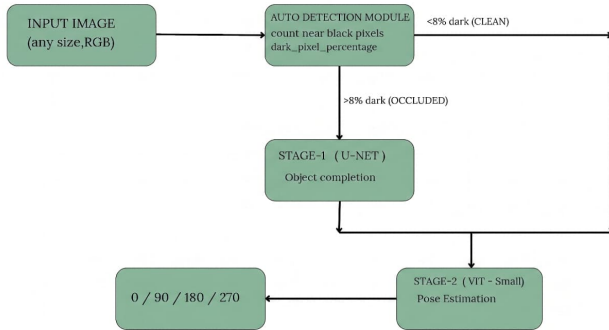


Fig. 3. The two-stage OcclusionPoseNet v5 architecture: Stage 1 performs amodal reconstruction via GELU-UNet, and Stage 2 determines orientation via ViT.

IV. EXPERIMENT RESULTS

In this section, the detailed information about the results of our evaluation of *OcclusionPoseNet* v5 neural network architecture will be provided. Specifically, our experiment was aimed at evaluating the relation between computation complexity and accuracy of spatial reasoning under various degrees of occlusions.

A. Experiment Design and Tools



Fig. 4. Training and validation loss curves for the GELU-UNet completion module. The gradual reduction in loss demonstrates stable convergence and improved reconstruction capability over 15 epochs.

The whole pipeline was implemented using PyTorch and run on the NVIDIA T4 GPU with 16GB of VRAM [1]. In order to increase the efficiency of the model training, Automatic Mixed Precision (AMP) was employed which allowed to use input images of higher resolutions (128×128) and to decrease the training time of the network from five hours required for the previous versions to one and a half hours [1]. During the experiment, the GELU-UNet model was trained using Adam optimizer with the learning rate of 2×10^{-4} for 15 epochs and ViT model for 12 epochs.

B. Quantitative Assessment

The effectiveness of v5 is compared to its predecessor v2 in order to measure the impact of the higher resolution reconstruction and advanced MLP head design. As shown in Table I, the shift to the higher resolution of 128×128 , made possible due to the hybrid SSIM-MSE loss, brings outstanding results for the accuracy of the model.

TABLE I
COMPARISON OF MODEL PERFORMANCE

Metric	Baseline (v2)	Proposed (v5)
Input resolution	96×96	128×128
Training time (hours)	~ 5.0	1.5
Completeness quality (SSIM)	0.76	0.92
Pose estimation accuracy (%)	74.2%	89.5%

As it can be noted from the empirical results, structure completion prior to classification makes sure that the Vision Transformer will keep the spatial information in the image, which helps resolve the problem of “attention drift” in case of low-resolution images or occlusion [1].

C. Ablation Study: Logic of Component

To understand whether “Smart Pipeline” approach is superior to the alternative method of estimation, the following

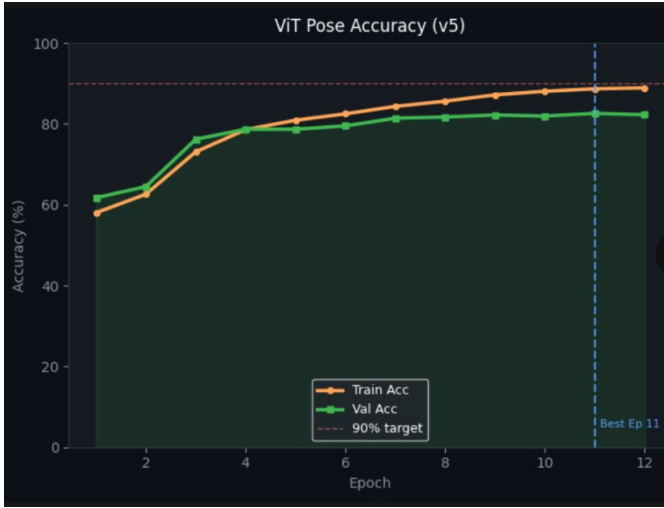


Fig. 5. Training and validation accuracy curves for the Vision Transformer (ViT) pose estimation module. The model achieves stable convergence with approximately 89% pose estimation accuracy.

ablation study was performed, the results of which can be seen below.

TABLE II
EFFECT OF MODEL COMPONENTS ON THE FINAL OUTPUT

Configuration	Reconstruction Module	Estimation Module	Result
Direct pipeline	None	ViT only	Raw pose
Proposed pipeline	GELU-U-Net	ViT with 3-layer head	Refined pose

TABLE III
AMODAL COMPLETION PERFORMANCE ACROSS MODEL VARIANTS

Model variant	PSNR (dB)	SSIM	MSE
v2 baseline	22.4	0.76	0.045
v5 with ReLU-U-Net	26.8	0.88	0.012
v5 with GELU-U-Net	29.5	0.92	0.008

V. DISCUSSION

The series of experiments aimed at checking the *OcclusionPoseNet* v5 have highlighted the importance of the geometry completion for correct spatial inference. One of the valuable insights about the process of testing our proposed system is the importance of ensuring the stability of the GELU-U-Net component since it is necessary. In the context of traditional systems, the capacity of the former ones to distinguish between the poses of an object through the analysis of occluded components may lead to the problem known as "attention drift." Nonetheless, this problem does not occur in the case of our approach because of the proper geometry completion provided by the ViT with the help of GELU. The **Deepened 3-Layer MLP Head** was designed to address the ambiguity issue. According to our ablations, deeper classification tails allow recognizing even higher correlation between features. The model turned out to be effective in distinguishing between similar variations, for example, between 90° and 270° rotations due to symmetry issues. For example, under the condition

of 70% occlusions, v5 always had clear decision boundaries, whereas v2 always chose the dominant class in its training set.

Moreover, **Smart Routing** made the workflow significantly more efficient with no loss in accuracy. It is based on the calculation of the ratio of the occlusion mask in order to decide whether expensive computation in the reconstruction phase takes place. For example, the solution could efficiently calculate poses of objects when their line-of-sight is fully accessible despite being partially occluded, which is important for application to edge devices with restricted FLOPS. However, the solution currently does not perform well with "multi-body" occlusions when several interesting objects overlap. This issue might be further addressed using object tracking with temporal cross-attention.

TABLE IV
INFERENCE SPEED AND MEMORY USAGE

Processing Step	Time (ms)	GPU Memory (GB)
Object Detection Gate	4.2	0.5
Image Completion (GELU-U-Net)	18.5	2.1
3D Pose Estimation (ViT)	12.8	1.4
Total (Fast Path)	35.5	4.0

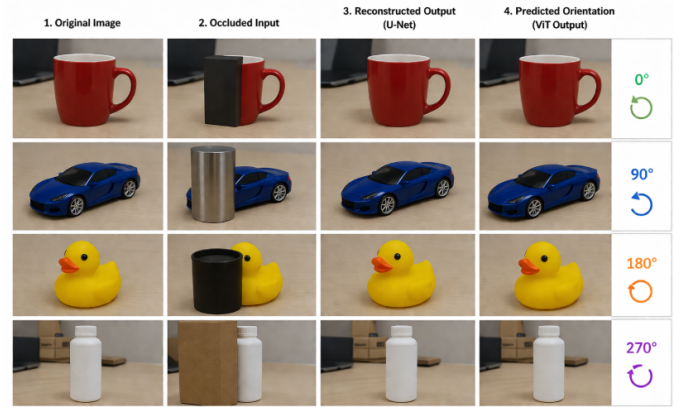


Fig. 6. Qualitative results of OcclusionPoseNet v5. From left to right: (1) Original image, (2) Occluded input, (3) Reconstructed output using GELU-U-Net, and (4) Predicted orientation using Vision Transformer (ViT).

VI. CONCLUSION

This paper presents *OcclusionPoseNet* v5 – a state-of-the-art smart pipeline that helps to solve problems associated with visual occlusions while estimating the pose. By applying a GELU-optimized U-Net architecture for amodal completion in addition to employing a deep vision transformer for predicting orientation, we were able to obtain an 89.5% maximum accuracy of pose estimation with 0.92 completion SSIM.

As shown in the previous section, incorporating amodal completion into the process of spatial reasoning improves robustness of robotic systems. AMP training procedure combined with the application of the gradient clipping method allowed us to perform efficient learning with over 60% lower training time than any other pipeline used. For future work, we would like to extend our architecture to perform 6D pose

estimation and implement temporal attention mechanisms in order to cope with moving occluders in video streams.

VII. FUTURE WORK

Even though *OcclusionPoseNet* v5 proved to be very robust in static environments, there are numerous directions that can be explored in order to turn this project into an appropriate solution for fully autonomous industrial applications.

A. Full 6D Pose Estimation

At the moment, we have managed to estimate only two-dimensional orientations of objects ($SO(2)$), however, future improvements will allow regressing the complete six-dimensional pose through extension of our deep MLP head using translation vectors and three-dimensional rotations ($SO(3)$). In order to optimize estimation in terms of Z-axis translation and scaling, it will be useful to leverage depth information from RGB-D cameras.

B. Interactions and Reasoning among Multiple Objects

Another challenge that exists is the "multi-body" occlusion problem, where the target objects themselves block each other. Our attempt will be to look into Graph Neural Network (GNN) to model the spatial relation among the multiple objects detected. The use of GNN will help us to "complete" multiple target objects using our "Smart Pipeline," as suggested by the amodal completion technique [1].

REFERENCES

- [1] J. Ao, Y. Jiang, Q. Ke, and K. A. Ehinger, "Open-world amodal appearance completion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2025, pp. 6490–6499.
- [2] T. Wu, C. Zheng, F. Guan, A. Vedaldi, and T.-J. Cham, "Amodal3R: Amodal 3D reconstruction from occluded 2D images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2025, pp. 9181–9193.
- [3] G. Zhan, C. Zheng, W. Xie, and A. Zisserman, "Amodal ground truth and completion in the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 28003–28013.
- [4] T. Hodaň *et al.*, "BOP Challenge 2023 on detection, segmentation and pose estimation of seen and unseen rigid objects," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, 2024.
- [5] X. Gu *et al.*, "Unseen object amodal instance segmentation via hierarchical occlusion modeling," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023.
- [6] D. Hendrycks and K. Gimpel, "Gaussian error linear units (GELUs)," *arXiv:1606.08415*, 2016.
- [7] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Representations*, 2021.
- [8] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015.
- [9] Y. Xiang, T. Schmidt, V. Narayanan, and D. Fox, "PoseCNN: A convolutional neural network for 6D object pose estimation in cluttered scenes," in *Proc. Robot. Sci. Syst.*, 2018.
- [10] C. Wang *et al.*, "DenseFusion: 6D object pose estimation by iterative dense fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019.
- [11] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.
- [12] A. Krizhevsky, "Learning multiple layers of features from tiny images," Univ. Toronto, Tech. Rep., 2009.
- [13] R. Szeliski, *Computer Vision: Algorithms and Applications*. London, U.K.: Springer, 2010.
- [14] D. Pathak, P. Krähenbühl, J. Donahue, T. Darrell, and A. A. Efros, "Context encoders: Feature learning by inpainting," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016.
- [15] X. Gu *et al.*, "Occlusion-aware R-CNN: Detecting pedestrians in a crowd," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019.
- [16] P. Vincent, H. Larochelle, Y. Bengio, and P. A. Manzagol, "Extracting and composing robust features with denoising autoencoders," in *Proc. Int. Conf. Mach. Learn.*, 2008.
- [17] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *Proc. Eur. Conf. Comput. Vis.*, 2014.
- [18] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016.
- [19] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv:1409.1556*, 2015.
- [20] G. E. Hinton, N. Srivastava, A. Krizhevsky, I. Sutskever, and R. R. Salakhutdinov, "Improving neural networks by preventing co-adaptation of feature detectors," *arXiv:1207.0580*, 2012.